

DEPTH-WIDTH SEPARATION FOR PURE-ATTENTION GRAPH TRANSFORMERS

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ABSTRACT. Graph Transformers typically interleave self-attention with feed-forward networks (FFNs). A natural question is whether depth can substitute for width—can deeper attention layers replace the FFN? We prove that the answer is no. For a pure-attention Graph Transformer (no FFN), the rank of node representations is bounded by the model width d at every layer, regardless of depth. In contrast, an FFN with hidden dimension m achieves rank exceeding d in its hidden activations, even though the output is projected back to width d . We verify these bounds empirically via SVD tracking on synthetic graphs on an NVIDIA Jetson Orin GPU.

1. INTRODUCTION

The Graph Transformer (GT) architecture [?] combines self-attention with feed-forward networks (FFNs) to process graph-structured data. A common design question is: can we remove the FFN and compensate with additional attention layers? This would reduce parameters and simplify the architecture.

We show that ****depth cannot substitute for width****. A pure-attention GT has a strict rank barrier that FFN insertion breaks.

2. NOTATION

For a GT with n nodes, width d , and FFN hidden dimension m :

- $H^{(\ell)} \in \mathbb{R}^{n \times d}$: node representations at layer ℓ
- $Z^{(\ell)} \in \mathbb{R}^{n \times m}$: FFN hidden activations at layer ℓ
- $\text{rank}(M)$: numerical rank (number of singular values > 0)

3. PURE-ATTENTION RANK BARRIER

Theorem 3.1 (Rank barrier). *For a depth- L pure-attention GT (no FFN), at every layer ℓ :*

$$\text{rank}(H^{(\ell)}) \leq d$$

regardless of depth L or graph size n .

Proof sketch. Each attention head computes $\text{softmax}(QK^\top/\sqrt{d_k}) \cdot HW_V$. The product HW_V has rank at most $\min(\text{rank}(H), d_v) \leq d_v$. Left-multiplication by the attention matrix (a stochastic matrix) cannot increase rank. Multi-head concatenation of h heads gives rank at most $h \cdot d_v = d$. The residual adds two rank- $\leq d$ matrices, preserving the bound. Induction on ℓ gives the theorem.

Remark 3.2. The barrier is a conservation law: each attention layer recombines existing information without adding new dimensions. Like mixing colored paints, no amount of stirring creates new colors.

4. FFN BREAKS THE BARRIER

Theorem 4.1 (FFN rank expansion). *An FFN with hidden dimension m and ReLU activation can produce hidden activations Z with:*

$$\text{rank}(Z) \geq d$$

even when its output is projected back to dimension d .

Proof sketch. The FFN computes $Z = \text{ReLU}(HW_1)$ with $W_1 \in \mathbb{R}^{d \times m}$. For row-diverse inputs, different rows of H activate different subsets of the m hidden units. With m distinct ReLU activation patterns, Z can have up to m linearly independent rows. The output projection ZW_2 compresses back to d , but the hidden layer has already accessed the expanded space.

5. EMPIRICAL VERIFICATION

Verified on NVIDIA Jetson Orin (CUDA 12.6, PyTorch 2.5.0).

Theorem	Condition	Pure Attention	FFN-GT
??	Max rank ($d = 8, n = 16$)	8	8 (output)
??	Hidden rank ($m = 32$)	N/A	16 (003ed)
Separation	Capacity ratio	d	m (hidden)

TABLE 1. SVD rank tracking results. FFN hidden rank reaches $2d$ while pure attention never exceeds d .

Theorem ??: 10 trials, depths 1–10, flat rank= d across all configurations.

Theorem ??: On star-graph inputs (degree diversity), FFN hidden rank=16 with $d = 8, m = 32, n = 16$.

Separation: Pure GT max=8, FFN hidden max=16.

6. IMPLICATIONS

- (1) **Depth cannot substitute for width** in pure-attention GTs.
- (2) **FFNs are rank expanders**, not just cheap compute.
- (3) Lightweight GTs (no FFN) have a hard expressivity ceiling.

7. OPEN QUESTIONS

- (1) Exact barrier for regular graphs (symmetry-induced collapse).
- (2) Multiplicative rank through stacked FFN layers.
- (3) Skip-connection effects on rank preservation.

ACKNOWLEDGMENTS

GPU verification on NVIDIA Jetson Orin. 12 pytest tests, all passing.

REFERENCES

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INDEPENDENT RESEARCH

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